Nguyễn Ngọc Minh, Kiều Sơn Tùng, Đặng Thịnh Tường Minh

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## Student information

- Nguyễn Ngọc Minh – 20235533
- Kiều Sơn Tùng – 20235571
- Đặng Thịnh Tường Minh – 20235528

## Presentation goal

Study whether a generalizable multi-view 3D reconstruction model can recover scene geometry from only a few overlapping images of a new scene, without per-scene optimization at test time.

- 3D reconstruction is important because 3D representations help machines understand spatial structure, depth, and object relationships more accurately.
- Practical 3D capture often has only a small number of usable images.
- Classical high-quality pipelines depend on accurate camera poses or expensive scene-specific optimization.
- Sparse views increase ambiguity because cross-view correspondence is harder to establish reliably.
- A fast, generalizable method for unseen scenes would be useful for robotics, AR/VR, navigation, and scalable environment digitization.

## Core question

How can we reconstruct unseen scenes accurately and efficiently when only 2 to 5 input views are available?

**Key constraints**

- Unseen scene at test time
- Sparse overlapping images
- No per-scene optimization

**Expected outcome**

- Robust geometry recovery
- Stable performance under view reduction
- Practical runtime per scene

1. Evaluate a generalizable 3D reconstruction model on unseen scenes.
2. Analyze sparse-view performance with 2, 3, and 5 images.
3. Compare a pairwise baseline (DUSt3R) with a single-stage multi-view method (MV-DUSt3R+).
4. Identify conditions where sparse-view reconstruction fails or remains reliable.

### Working hypothesis

A feed-forward multi-view model should be more robust than a pairwise baseline when the number of input views is limited, because it reasons over multiple views jointly in one pass.

- Better global consistency across views
- Less dependence on separate alignment stages
- Stronger behavior in highly ambiguous sparse-view settings

## Our contribution

Beyond benchmarking DUSt3R and MV-DUSt3R+, we first identify which sparse-view components help, then propose a supervised extension for the failure cases that heuristic filters could not solve.

- **View selection for far-reference issue:** replace random sparse-view sampling with policies that balance scene coverage, viewpoint diversity, and overlap.
- **Honest ablation of heuristic fusion/filtering:** test confidence, multi-view support, occlusion, and ambiguity signals separately.
- **Supervised reliability extension:** replace failed hand-written filters with learned occlusion reliability and repeated-structure match disambiguation heads.

## 1. View selection for far-reference issue

- Compare **random baseline**, **coverage-aware**, **diversity-aware**, and **hybrid** sparse-view policies.
- The goal is to reduce far-reference failures while preserving co-visibility under 2–5 input views.

## 2. Heuristic ablation before adding learned modules

- Evaluate $C$, $S_{mv}$, $M$, occlusion penalties, and ambiguity penalties independently.
- If a module reduces F-score or completeness, keep it out of the final pipeline and report the failure explicitly.

## 3. Supervised occlusion and ambiguity heads

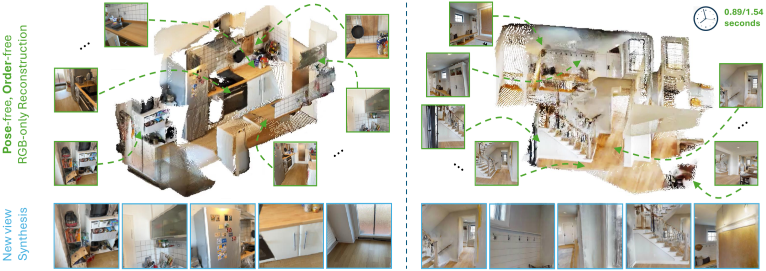
- Train an **Occlusion-Aware Reliability Head** to keep occluded-but-valid points and reject floating/wrong geometry.
- Train a **Repeated-Structure Disambiguation Head** using reciprocal matches, descriptor margin, reprojection, and cycle error.

## Why not keep tuning heuristic filters?

Early ablations show that hand-written occlusion and ambiguity filters can remove valid geometry and reduce completeness. The next step is supervised reliability learning rather than stronger manual penalties.

- **OARH:** predict whether each candidate point should be kept, using confidence, visibility, occlusion, reprojection, depth disagreement, and multi-view agreement features.
- **RSDH:** predict whether repeated-structure matches are valid, using MASt3R reciprocal matching, feature margin, 3D disagreement, and cycle consistency.
- **Fine-tuning policy:** freeze MV-DUSt3R+ first; only later unfreeze the confidence head and last decoder blocks if validation improves.

- Primary method: **MV-DUSt3R+**
- Single-stage feed-forward multi-view reconstruction
- Targeted at sparse-view scenes without camera calibration
- Our study focuses on robustness, quality, and runtime




Source: Tang et al., MV-DUSt3R+, project page and paper

# DUSt3R baseline

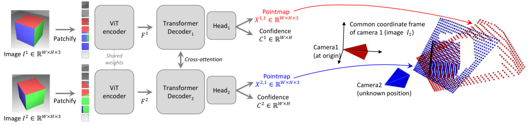
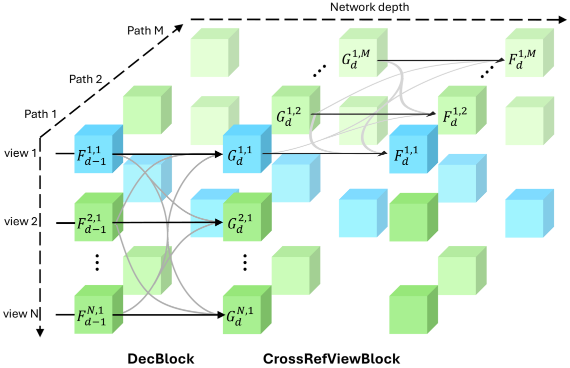



Figure 2. **Architecture of the network.** Two views of a scene $(I^1, I^2)$ are first encoded in a Siamese manner with a shared ViT encoder. The resulting token representations $F^1$ and $F^2$ are then passed to two transformer decoders that constantly exchange information via cross-attention. Finally, two regression heads output the two corresponding pointmaps and associated confidence maps. Importantly, the two pointmaps are expressed in the same coordinate frame of the first image $I^1$. The network is trained using a simple regression loss (Eq. (4))

Pairwise reasoning with cross-attention between two views

# MV-DUSt3R+ main method



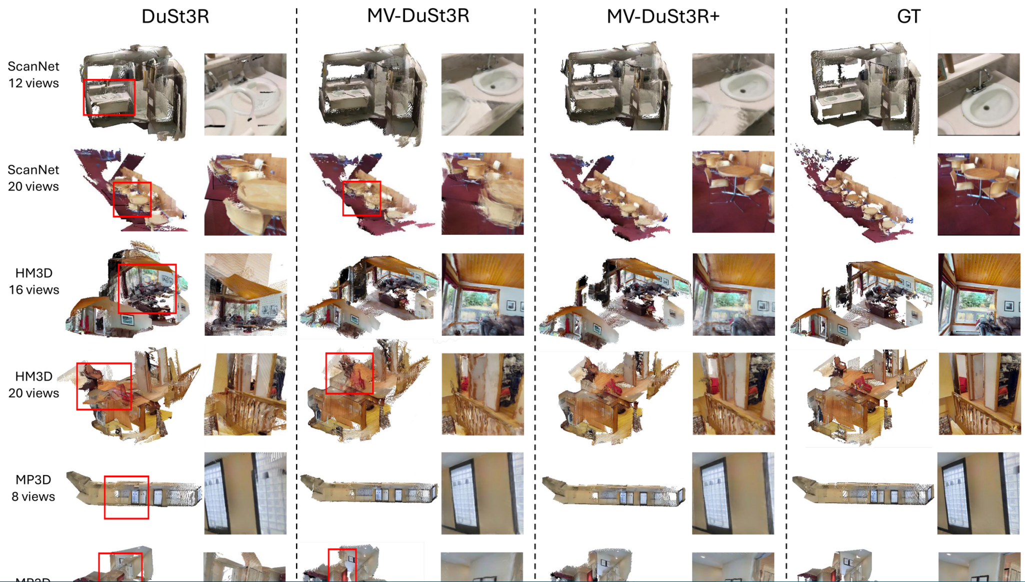

Multi-path design with cross-reference-view information flow

Source: Left: Wang et al., DUSt3R, CVPR 2024. Right: Tang et al., MV-DUSt3R+, project page/paper

| Aspect | DUSt3R | MV-DUSt3R+ |
|---|---|---|
| Processing style | Pairwise reconstruction | Joint multi-view reconstruction |
| Global consistency | Requires later alignment | Built into one forward pass |
| Sparse-view behavior | Can accumulate pairwise errors | More robust cue fusion |
| Runtime trend | Slower as views increase | Better scaling with more views |
| Role in project | Reference baseline | Main candidate method |

- DUSt3R is still a strong baseline, so the comparison is direct and meaningful.
- MV-DUSt3R+ is better aligned with our target setting of 2–5 sparse views.

| | Method | GO | HM3D | | | ScanNet | | | MP3D | | | Time |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | ND ↓ | DAc ↑ | CD ↓ | ND ↓ | DAc ↑ | CD ↓ | ND ↓ | DAc ↑ | CD ↓ | (sec) |
| 4 views | Spann3R | × | 37.1 | 0.0 | 225(184) | 8.9 | 19.5 | 54.7(50.1) | 42.7 | 0.0 | 248(202) | 0.36 |
| | DUSt3R | ✓ | 1.9 | 75.1 | 5.6(2.3) | 1.3 | 89.8 | 4.0(0.4) | 3.9 | 41.7 | 40.0(5.3) | 2.42 |
| | MV-DUSt3R | × | 1.1 | 92.2 | 2.0(1.1) | 1.0 | 93.3 | 2.0(0.4) | 2.5 | 62.4 | 25.3(4.1) | 0.05 |
| | MV-DUSt3R+ | × | 1.0 | 95.2 | 1.5(0.9) | 0.8 | 94.9 | 1.5(0.3) | 2.2 | 68.0 | 19.9(3.4) | 0.29 |
| | MV-DUSt3R$_{oracle}$ | × | 1.0 | 94.6 | 1.5(0.7) | 0.8 | 95.5 | 1.3(0.3) | 2.3 | 66.6 | 20.7(4.0) | - |
| | MV-DUSt3R+$_{oracle}$ | × | 0.9 | 96.5 | 1.4(0.7) | 0.7 | 95.8 | 1.2(0.2) | 2.1 | 70.6 | 17.9(3.3) | - |
| 12 views | Spann3R | × | 32.6 | 0.0 | 125(113) | 9.1 | 16.3 | 36.6(31.2) | 35.0 | 0.0 | 138(112) | 1.34 |
| | DUSt3R | ✓ | 3.9 | 30.7 | 18.1(3.4) | 1.9 | 82.6 | 4.1(0.6) | 6.6 | 12.0 | 49.6(8.3) | 8.28 |
| | MV-DUSt3R | × | 1.6 | 79.5 | 3.0(1.2) | 1.4 | 86.8 | 2.3(0.8) | 3.4 | 41.3 | 22.6(5.5) | 0.15 |
| | MV-DUSt3R+ | × | 1.2 | 91.5 | 1.8(0.7) | 1.2 | 88.4 | 1.8(0.7) | 2.6 | 55.0 | 15.1(3.8) | 0.89 |
| | MV-DUSt3R$_{oracle}$ | × | 1.3 | 88.8 | 1.8(0.9) | 1.0 | 90.6 | 1.3(0.7) | 2.9 | 51.3 | 16.4(4.0) | - |
| | MV-DUSt3R+$_{oracle}$ | × | 1.1 | 94.8 | 1.4(0.7) | 1.0 | 90.9 | 1.3(0.5) | 2.5 | 59.8 | 13.6(3.5) | - |
| 24 views | Spann3R | × | 41.7 | 0.0 | 139(121) | 11.4 | 1.6 | 37.4(35.5) | 46.6 | 0.0 | 151(121) | 2.73 |
| | DUSt3R | ✓ | 6.8 | 7.3 | 32.4(5.2) | 2.4 | 72.6 | 5.1(1.0) | 11.4 | 2.5 | 80.9(14.3) | 27.21 |
| | MV-DUSt3R | × | 3.4 | 36.7 | 10.0(3.5) | 2.2 | 75.2 | 2.7(0.9) | 6.3 | 12.2 | 38.6(13.9) | 0.35 |
| | MV-DUSt3R+ | × | 2.1 | 64.5 | 3.9(2.0) | 1.6 | 81.2 | 1.7(0.7) | 4.3 | 26.7 | 22.0(5.9) | 1.97 |
| | MV-DUSt3R$_{oracle}$ | × | 2.1 | 58.9 | 3.5(2.1) | 1.4 | 82.9 | 1.4(0.7) | 4.4 | 22.0 | 19.9(5.1) | - |
| | MV-DUSt3R+$_{oracle}$ | × | 1.8 | 77.9 | 2.6(1.3) | 1.3 | 85.1 | 1.3(0.6) | 3.6 | 33.1 | 15.1(4.4) | - |


DuSt3R
MV-DuSt3R
MV-DuSt3R+
GT
ScanNet
12 views
ScanNet
20 views
HM3D
16 views
HM3D
20 views
MP3D
8 views

Prepare multi-view scenes and split into train / validation / unseen test

↓

Create sparse-view subsets with 2, 3, 4, and 5 input images

↓

Run MV-DUSt3R+ pretrained with selected sparse views

↓

Apply learned reliability / visibility / ambiguity heads

↓

Evaluate geometry, robustness, runtime, and case-specific failures

**Controlled factors**

- Number of input views
- Scene-level train / validation / test split
- Fixed threshold on final test

**Observed outcomes**

- Reconstruction accuracy
- Completeness and consistency
- Runtime per scene
- Occlusion and repeated-structure failures

1. First build a supervised label cache with per-view visibility, occlusion, floating/wrong-depth, and match labels.
2. Freeze MV-DUSt3R+ and train small reliability / match heads on generated GT labels.
3. Evaluate OARH on occlusion-heavy scenes and RSDH on repeated-structure scenes before mixing them into the full pipeline.
4. If validation improves, unfreeze only the confidence head and last decoder blocks for light fine-tuning.
5. Keep final test clean: no threshold tuning, no checkpoint selection on test scenes.

## Primary dataset: ScanNet++

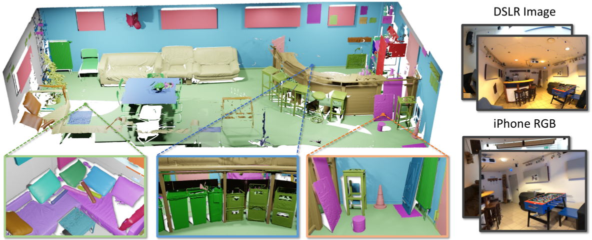
High-fidelity indoor scenes with registered DSLR images and commodity iPhone captures.

## Secondary dataset: DTU

Controlled object-centric scenes for supplementary evaluation and sanity checks in a cleaner setting.

If compute is limited, we will use curated subsets rather than the full datasets.

Source: Yeshwanth et al., ScanNet++, ICCV 2023, Figure 1



| Dataset | Num. scenes | Average scans/scene | Total scans | RGB (MP) | | Depth | | Dense semantics | NVS |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Commodity | DSLR | LR | HR | | |
| LLFF [26] | 35 | - | - | 12 | ✗ | ✗ | ✗ | ✗ | ✓ |
| DTU [15] | 124 | - | - | 1.9 | ✗ | ✗ | ✓ | ✗ | ✓ |
| BlendedMVS [47] | 113 | - | - | ✗ | ✗ | ✗ | ✗ | ✓ | ✗ |
| ScanNet [9] | 1503 | - | - | 1.25 | ✗ | ✓ | ✗ | ✓ | ✗ |
| Matterport3D [4] | 90‡ | - | - | 1.3 | ✗ | ✓ | ✗ | ✓ | ✗ |
| Tanks and Temples [20] | 21 | 10.57 | 74 | ✗ | 8 | ✗ | ✓ | ✗ | ✓ |
| ETH3D [35] | 25 | 2.33 | 42 | ✗ | 24 | ✗ | ✓ | ✗ | ✗ |
| ARKitScenes [3] | 1004 | 3.16 | 3179 | 3 | ✗ | ✓ | ✓ | ✗ | ✗ |
| *ScanNet++* (ours) | 460 | 4.85 | 1858 | 2.7 | 33 | ✓ | ✓ | ✓ | ✓ |

Source: Yeshwanth et al., ScanNet++, ICCV 2023, Table 1

- 460 indoor scenes with high-fidelity geometry
- DSLR images support stronger geometric supervision
- Commodity captures make sparse-view testing realistic
- Dense semantics and NVS setup make the benchmark richer

- Reconstruction accuracy against available ground-truth geometry
- Completeness and consistency of reconstructed point clouds
- Performance degradation under 2-view, 3-view, and 5-view settings
- Inference efficiency, including runtime per scene

| | Method | GO | HM3D | | | ScanNet | | | MP3D | | | Time |
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**Table 2 MVS reconstruction results.** Best results are marked in red. For clarity, metric ND is scaled up by 10×, DAc is in percent %, and CD is scaled by 100× with its median given in parentheses. Results of our **oracle** methods are obtained by manually selecting the best single reference view from reference view candidates to report a possibly achievable

## Success criterion

Show that the multi-view method is more robust than the pairwise baseline under identical sparse-view conditions.

Source: Tang et al., MV-DUSt3R+, quantitative benchmark

- End-to-end implementation for training and evaluation
- Quantitative results on unseen scenes across multiple view counts
- Qualitative 3D visualizations and failure-case analysis
- Final report discussing strengths, limitations, and recommendations
- Reproducible Kaggle experiment scripts organized under `scripts/kaggle`
- Final validation rerun script with a P100-compatible Torch fallback for unstable Kaggle GPU allocation
- Submitted supervised-extension kernels for Run 12–18, including reliability, match-disambiguation, reciprocal-feature, fine-tune-gate, and learned-summary runs
- Submitted Phase 3 label-cache kernel for Run 19 to support stricter occlusion and ambiguity supervision
- Clean GitHub repository with curated docs, proposal sources, and PDF deck

- A focused benchmark of sparse-view reconstruction on unseen scenes
- A direct comparison between pairwise and single-stage multi-view reasoning
- Evidence about when sparse-view reconstruction remains reliable
- Practical guidance for follow-up work on robust 3D reconstruction
- An ablation record showing which proposed modules help and which fail
- A supervised follow-up plan with honest outcomes: gated OARH still needs better labels/features, while RSDH is strong on a GT-depth-assisted MASt3R proxy but not yet a final image-only ambiguity solution
- A stricter Phase 3 protocol that requires held-out occlusion-heavy and repeated-structure gains before claiming the remaining limitations are solved

- Wang et al., "DUSt3R: Geometric 3D Vision Made Easy," CVPR 2024.
- Tang et al., "MV-DUSt3R+: Single-Stage Scene Reconstruction from Sparse Views In 2 Seconds," CVPR 2025.
- Yeshwanth et al., "ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes," ICCV 2023.
- DTU Robot Image Data Sets, Multi-View Stereo Benchmark.